TfL Delay Intelligence System

Real-Time Anomaly Detection for Urban Transit
Real-Time ML System for Transit Delay Monitoring

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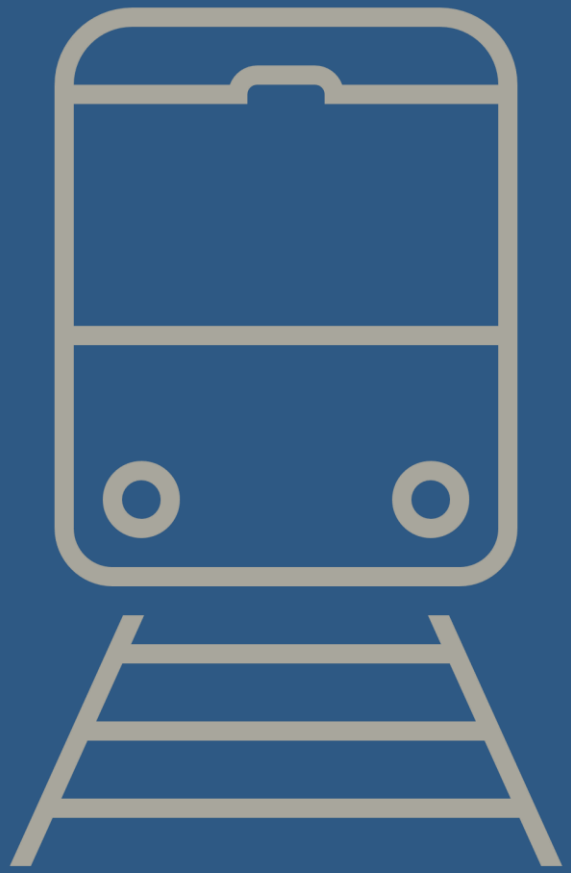

Capstone Project Pitch | March-April 2026 Time-Aware Classification & Intelligent Monitoring

Real-World Problem & Our Data Engineering Solution

The Problem

- Passengers rely on arrival predictions like "Train arriving in 2 minutes," but these systems do not highlight emerging abnormal delay patterns
- Need to detect abnormal delay patterns before they become obvious

Our Data Engineering Approach

- Real-time API ingestion: TfL StopPoint Arrivals API every 30 seconds
- Live data collection: SQLite with WAL mode for safe concurrent access
- Prediction stream capture: ~900k arrival prediction records per 6-hour window from Victoria & Jubilee lines

End-to-End ML Pipeline: From Real-Time Data to Intelligence

1. **Real-Time Ingestion**
   TfL API every 30s
   900k+ records/6hrs

2. **Safe Storage**
   SQLite WAL Backup snapshots

3. **Feature Engineering**
   Rolling windows
   Seasonal baselines

4. **ML Training**
   Time-aware splits
   PR-AUC evaluation

5. **Risk Scoring**
   Anomaly detection
   Real-time alerts

Real-Time Data Engineering Scale

30s
Collection Interval

900k+
Records per 6 hours

2
Tube Lines

24M+
Records in 192 hours or ~8 days

Modern ML Pattern

Real-time anomaly classification Same approach as fraud detection, cybersecurity monitoring, and industrial IoT systems

Not Traditional Forecasting

Focus on anomaly detection We classify current behavior using time-aware features rather than forecasting future arrival times

Example System Output

Real-Time Anomaly Detection in Action

Tuesday Morning Rush Hour - Oxford Circus Station

Station Context

Station Oxford Circus	**Line** Victoria
Time 8:30 AM	**Day** Tuesday

Anomaly Risk: 0.82

HIGH RISK DETECTED

Current prediction significantly exceeds normal baseline for this station, time, and day. Rolling window analysis shows sustained delay pattern over past 10 minutes.

Current vs Historical

Current Prediction:	11 minutes
Normal Baseline:	4 minutes
Deviation:	+7 minutes
Rolling Mean (10m):	9.2 minutes

System Insights

- **Delay trend increasing**
- **175% above normal**
- **Pattern sustained 10+ min**
- **Early warning triggered**

Key Value Proposition

We detect abnormal delay patterns 7+ minutes before they become obvious, enabling proactive passenger communication and resource allocation during emerging disruptions.

Technical Architecture & Production Stack

1 Data Engineering

Real-time API ingestion with safe storage and backup systems

TfL API | SQLite WAL | Parquet | Python

- 900k+ data points every 6 hours
- WAL mode + backup snapshots
- 30-second collection intervals
- Structured ETL pipeline

2 Feature Engineering

Time-aware features with rolling windows and seasonal baselines

Pandas | NumPy | Rolling Windows | Temporal Features

- 10-minute rolling statistics
- Seasonal baselines (hour + weekday)
- Time-aware validation splits
- Multi-resolution temporal features

3 ML & Intelligence

Multiple algorithms with ensemble methods and explainable AI

Scikit-learn | XGBoost | SHAP | PR-AUC

- Ensemble classification methods
- Sub-minute inference latency
- SHAP explainability layer
- Imbalanced data optimization

Production ML Principles

- Data Safety & Validation
- Scalable Architecture
- Real-time Processing
- Explainable AI
- System Monitoring
- Model Versioning

ML Algorithms to Explore

- Random Forest
- XGBoost
- Logistic Regression
- Gradient Boosting
- Ensemble Methods
- Hyperparameter Tuning

Business Impact & Project Timeline

Quantified Business Value

£70-90M

Annual value in London alone*

*Based on research estimating passenger time savings from improved real-time transit information

- Reduces passenger uncertainty and waiting times
- Enables proactive resource allocation
- Encourages sustainable transport adoption

Technical Achievements

- **Data Engineering:** Real-time API ingestion pipeline
- **ML Engineering:** Time-aware classification patterns
- **System Design:** Sub-minute latency architecture
- **Production Ready:** Data safety & monitoring standards

4-Week Implementation Roadmap (Target: April 7 Demo)

Week 1

Data collection pipeline
Feature engineering
Baseline model

Week 2

Advanced modeling
Performance optimization
Evaluation framework

Week 3

Real-time inference
Streamlit demo app
System integration

Week 4

Final testing
Documentation
Presentation prep

This project demonstrates modern ML engineering: real-time data systems, production-ready architecture, and quantifiable business impact.